



Zoltrix Nightingale PRO 6	Intel 845GBV †	MSI 845 MAX2 BLR †
CM18738-6CH	AD1981A	AL650
5.1	2	5.1
✗	✗	✗
DirectSound, DS3D, EAX 1.0, EAX Advanced HD, EAX 2.0, and A3D 1.0	DirectSound, DS3D, EAX 1.0, EAX 2.0, I3DL2	DirectSound, DS3D, EAX 1.0, EAX Advanced HD, EAX 2.0
✗	✗	✗
Windows 9x/NT/2K/XP	Windows 3.x/9x/NT/2K/XP	Windows 3.x/9x/NT/2K/XP
Wavetable Based on System RAM 2, 4, 8 128, NA 48	Wavetable Based on System RAM NA NA 16	Wavetable Based on System RAM NA NA 48
✗	✗	✓
Full 16 bit @ 48 KHz 16 bit @ 48 KHz 85	Full 16 bit @ 48 KHz 16 bit @ 48 KHz 90	Full 16 bit @ 48 KHz 16 bit @ 48 KHz 100
✓	✓	✓
✓	✓	✓
✓	✓	✓
✓	✗	✓
✓	✗	✓
✗	✗	✓
✓	✗	✗
✓	✗	✗
✓	✗	✗
✗	✗	✗
✓	✗	✓
✗	✓	✓
✗	✗	✓
2.2	2.0	1.8
4.0	1.7	2.8
48.1 / 48.8	47.4 / 48.8	32.9 / 32.9
7.71	1.00	2.53
-5.81, +0.09	-0.03, +0.11	-0.42, +0.24
-67.2	-85.8	-80.8
67.9	83.8	78.5
6.997	0.0122	0.0022
-68.1	-79.2	-79.2
8	8	15
2	10	10
1	1	1
20.50	10.00	22.00
26.77	23.05	22.77
4.00	5.00	6.00
14.77	20.00	20.00
66.04	58.05	70.77
A-	B	A-
Kunhar Peripherals Pvt Ltd	Nebula Technologies	Priya Limited
salesbom@ kunhar.com	pankaj@ nebulatech.com	salesbom@priyagroup.com
2,000	8,120 (for motherboard)	10,600 (for motherboard)

that the overhead caused by the soundcard would not affect your work.

Technical test (RMAA): RightMark Audio Analyzer was used to test the core circuitry of the soundcard and to determine its performance. It is a loop-back test, wherein sound generated by the card is sent through the Line Out jack and is then reverted back to the card from the Microphone In or Line In port. This process effectively removes bottlenecks such as a slow processor or the presence of older peripherals. RMAA thus tests the raw soundcard performance. We tested the soundcards for five varied set of parameters:

a) Frequency response: This checks whether the soundcard distorts the relative loudness of different frequencies in a sound signal. Errors in frequency response are generally thought to be the most audible form of linear distortion. The Extigy almost touched perfection, generating a minimal difference of 1.58 dB. It was closely followed by the Creative Vibra 128 with a difference of 2.45 dB. The Creative Sound Blaster Live! DE 5.1 was surprisingly at the bottom with a frequency response of 26.37 dB (range of -4.81, +21.56), indicating that on the higher pitch the soundcard produces a lot of distortion.

b) Signal-to-Noise level: This test estimates the noise level that can be heard when nothing is playing and the volume level on the speakers is turned up high. It also helps to locate interference caused by electronic circuits, CRT displays and other such devices. The results here are in the negative and a higher number means better performance under this test. The Creative Sound Blaster Extigy was at the top of the pool. With an excellent attenuation of -104.3 dB, this card hardly produces any noise. The Vibra 128 and Zoltrix Nightingale PRO 6 generated really poor scores of -60.7 dB and -67.2 dB respectively, which indicated that a lot of disturbance and humming were produced by these soundcards. The rest performed adequately with an average attenuation of -80 dB.

c) Dynamic range: The dynamic range of a card determines the range of frequencies within which it can produce clean sound. This test also estimates the noise level in the presence of a weak audio signal and sheds light on the limit of the sound device while operating at low signal levels—an important aspect for high-quality sound recording and playback.

The results were quite as expected with the Vibra 128 remaining at the bottom of the ladder with a score of 58.3 dB while the Extigy scored 92.5 dB! The Extigy is therefore an excellent solution for professional quality audio recording and playback.

d) Total Harmonic Distortion (THD) and Noise: This test estimates the amount of harmonic distortion that occurs when an audio signal with large amplitude passes the testing chain. THD is determined by measuring the size of each of the new



Amplified Soundcards

Older ISA cards had an on board amplifier and did not require an amplified speaker set. Due to the distortion in sound that this practise caused, newer soundcards do not come with amplifiers. So, if your new soundcard does not play anything on your old speakers, you might have a non-amplified set that will need to be replaced.